

PAN-CANCER MBD4 LOSS-OF-FUNCTION: CYTIDINE + ATRi

MBD4-LOF Dual Therapeutic Framework: Cytidine + ATRi

MBD4 Loss-of-Function

**BER Pathway Deficiency
(Glycosylase loss)**

**Unprocessed U:G Mismatches
at CpG Sites**

*BER substrate
accumulation*

**Accumulated U:G
Mismatch Substrate**

**Gemcitabine / Cytarabine
Exploit BER Substrate**

**AXIS 1: CYTIDINE ANALOGS
(Validated — isogenic + rescue)**

*No PARP1 Upregulation
(No Selective Transcriptional Induction)*

*Constitutive
fork stalling*

**Fork-Blocking Lesions
→ Replication Stress**

**ATR Checkpoint
Dependency**

**AXIS 2: ATR INHIBITION
(p=0.003, d=-0.74; 4/4 stress)**



REPLICATION FORK COLLAPSE